

Homework

#1: Design a fully bridge rectifier

#2: Design an active rectifier with only a switch to clear residual charge

Deliverable: A report giving the schematic, waveform, output power, and some explanations.

Homework #1

Designing a full-bridge rectifier

PT parameters:

$$I_P = I_0 * \sin 2\pi f$$

Where $I_0 = 50 \mu\text{A}$

$f = 200 \text{ Hz}$

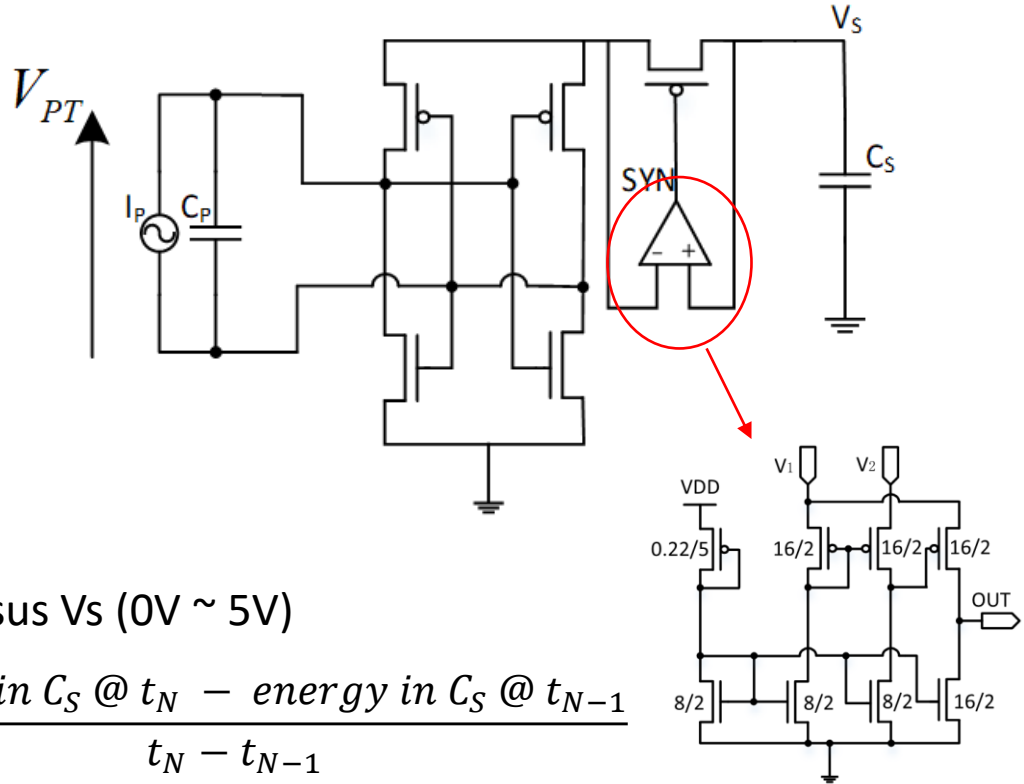
$C_P = 10 \text{ nF}$

C_S between $1 \mu\text{F}$ and $100 \mu\text{F}$

Deliverables:

- Simulated V_{PT} waveform.
- Simulated figure of Power versus V_S ($0\text{V} \sim 5\text{V}$)

$$\text{Power @ } t_N = \frac{\text{Energy in } C_S \text{ @ } t_N - \text{energy in } C_S \text{ @ } t_{N-1}}{t_N - t_{N-1}}$$



Homework #2

Designing a zero-stage SSHC rectifier

PT parameters:

$$I_P = I_0 * \sin 2\pi f$$

Where $I_0 = 50 \mu\text{A}$

$f = 200 \text{ Hz}$

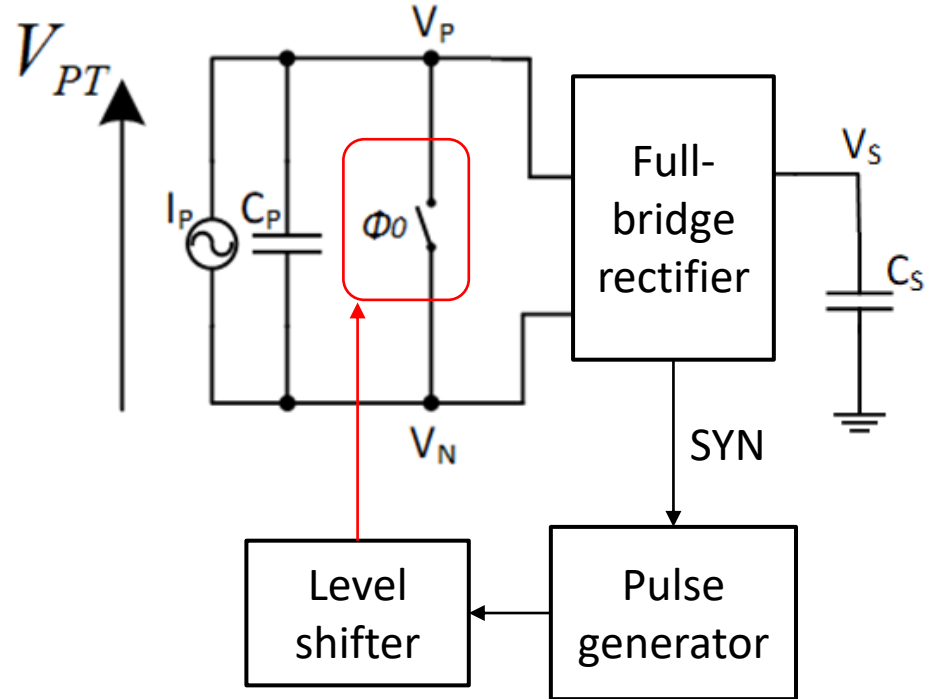
$C_P = 10 \text{ nF}$

C_S between $1 \mu\text{F}$ and $100 \mu\text{F}$

$V_{DD} = 1.5 \text{ V}$

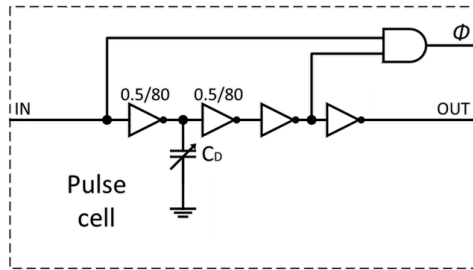
Deliverables:

- Simulated V_{PT} waveform showing voltage flipping
- Simulated figure of Power versus V_S ($0\text{V} \sim 5\text{V}$)
- Compare your power figure with HW #1.



Homework #2

Pulse generator



Level shifter

